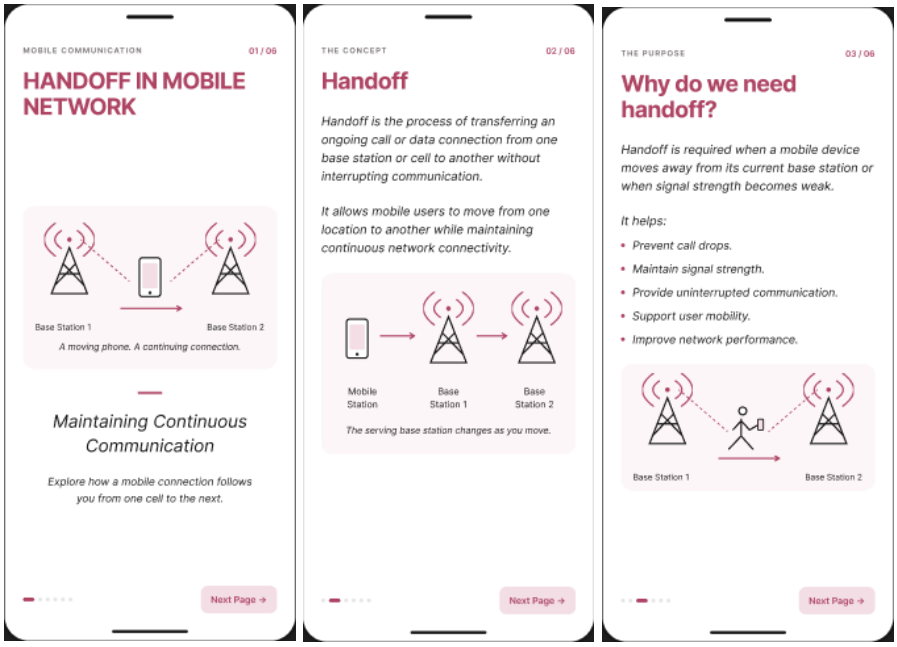
Exp-6: Use Figma to simulate a handoff scenario in a mobile network and visualize the process.

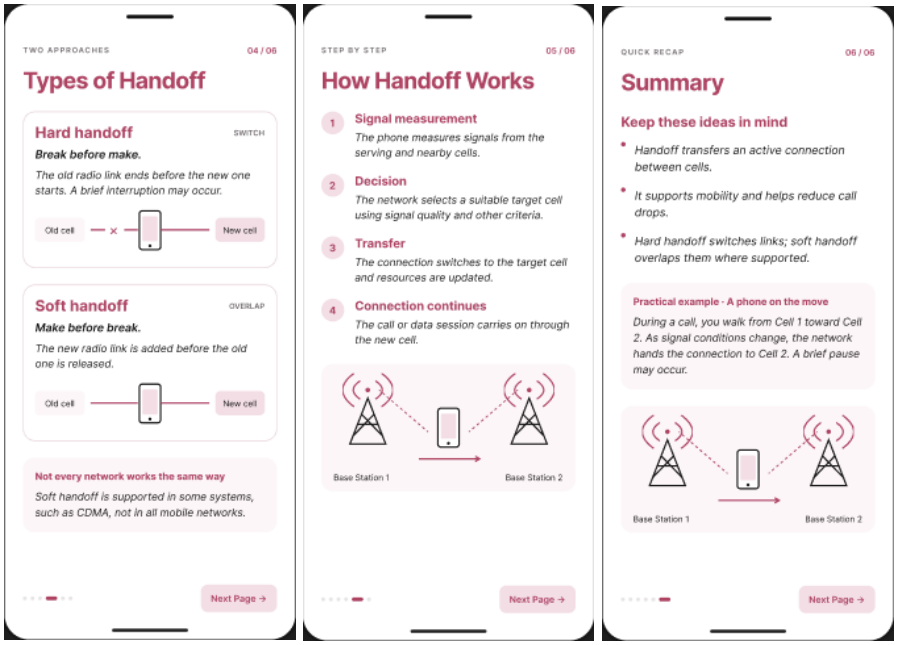
Aim: To simulate a handoff scenario in a mobile network and visualize the process using Figma.

Procedure:

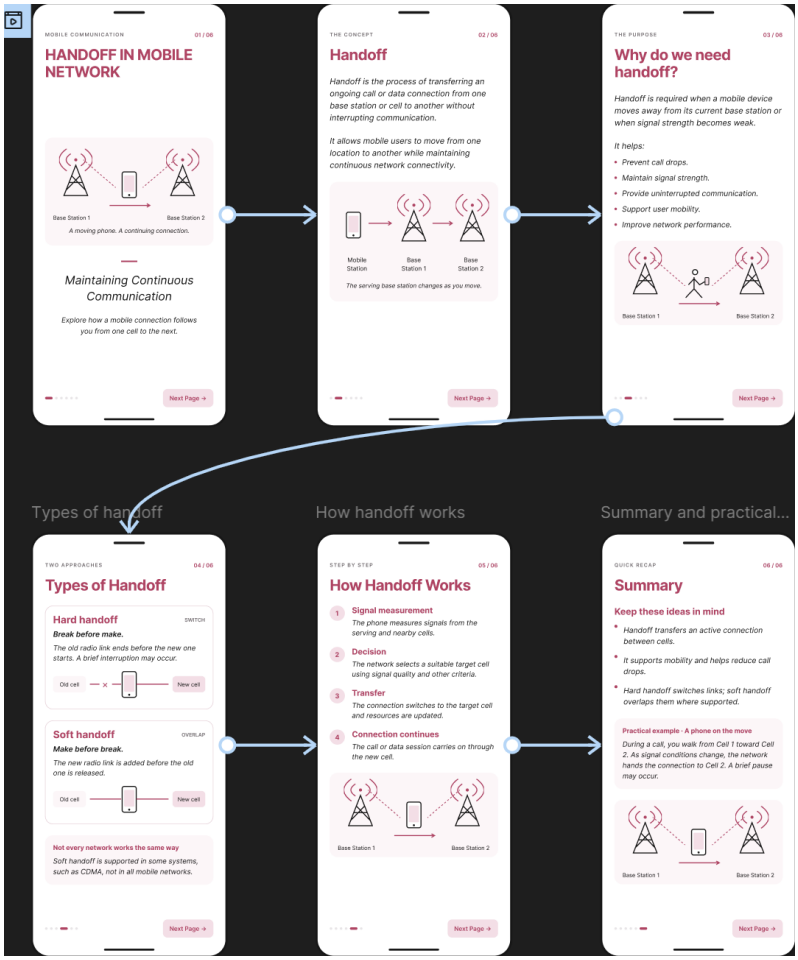
1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
10. Review and edit the Prototype
11. Save and Share

Design:




TWO APPROACHES
04 / 06
Types of Handoff
Hard handoff
SWITCH
Break before make.
The old radio link ends before the new one starts. A brief interruption may occur.
Old cell
New cell
Soft handoff
OVERLAP
Make before break.
The new radio link is added before the old one is released.
Old cell
New cell
Not every network works the same way
Soft handoff is supported in some systems, such as CDMA, not in all mobile networks.
Next Page →
STEP BY STEP
05 / 06
How Handoff Works
1 Signal measurement
The phone measures signals from the serving and nearby cells.
2 Decision
The network selects a suitable target cell using signal quality and other criteria.
3 Transfer
The connection switches to the target cell and resources are updated.
4 Connection continues
The call or data session carries on through the new cell.
Base Station 1
Base Station 2
Next Page →
QUICK RECAP
06 / 06
Summary
Keep these ideas in mind
Handoff transfers an active connection between cells.
It supports mobility and helps reduce call drops.
Hard handoff switches links; soft handoff overlaps them where supported.
Practical example - A phone on the move
During a call, you walk from Cell 1 toward Cell 2. As signal conditions change, the network hands the connection to Cell 2. A brief pause may occur.
Base Station 1
Base Station 2
Next Page →

Prototype:


MOBILE COMMUNICATION
01 / 06
HANDOFF IN MOBILE NETWORK
Base Station 1
Base Station 2
A moving phone. A continuing connection.
Maintaining Continuous Communication
Explore how a mobile connection follows you from one cell to the next.
Next Page →
THE CONCEPT
02 / 06
Handoff
Handoff is the process of transferring an ongoing call or data connection from one base station or cell to another without interrupting communication.
It allows mobile users to move from one location to another while maintaining continuous network connectivity.
Mobile Station
Base Station 1
Base Station 2
The serving base station changes as you move.
Next Page →
THE PURPOSE
03 / 06
Why do we need handoff?
Handoff is required when a mobile device moves away from its current base station or when signal strength becomes weak.
It helps:
Prevent call drops.
Maintain signal strength.
Provide uninterrupted communication.
Support user mobility.
Improve network performance.
Base Station 1
Base Station 2
Next Page →
Types of handoff
How handoff works
Summary and practical...
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Result:

Hence the simulated a handoff scenario in a mobile network and visualized the process using Figma.